Physics 9th – Chapter-8-Magnetism (Short Response Questions)

Q1. Can two magnetic field lines intersect each other? Justify your answer.

Ans:

- No, two magnetic field lines can never intersect.
- The direction of the magnetic field at a point is the direction of force on a north pole placed there, so it has only one direction.
- If lines crossed, there would be two directions at one point, which is not possible.

Q2. A freely suspended magnet always points along north-south direction. Why?

Ans:

- The Earth is a big magnet with its own magnetic field.
- The north pole of the suspended magnet is attracted towards the Earth's magnetic pole in the north, and its south pole is attracted to the opposite one.
- Because unlike poles attract, the magnet settles in the north-south direction.

Q3. What is the neutral zone or field free region of the magnetic field?

Ans:

- A neutral zone is a region where the magnetic field is zero.
- It is found by placing like poles (for example two N-poles) side by side, so their fields cancel each other.
- This idea is used in magnetic shielding.

Q4. Is there any material which does not have any magnetic behavior? Justify your answer.

Ans:

- Yes, non-magnetic (diamagnetic) materials have no magnetic behavior.
- In their atoms the spin and orbital motions of electrons cancel each other, so the net magnetic field is zero. They are not attracted by a magnet (only very slightly repelled).
- Examples are rubber, plastic, wood, glass and copper.

Q5. A proton is also a charged particle and spins like an electron. Why is its effect neglected in the study of magnetism?

Ans:

- The magnetic moment is $M = eh/4\pi m$, so $M \propto 1/m$.
- Since the mass of a proton is 1837 times the mass of an electron ($mp = 1837\ me$), the magnetic moment of a proton is much smaller than that of an electron ($Mp << Me$).
- So its effect is neglected.

Q6. What is the geomagnetic reversal phenomenon? Explain.

Ans:

- In geomagnetic reversal, the Earth's magnetic field flips, so the magnetic north and south poles change places.
- It happens slowly over thousands of years, during which the field becomes weaker.

- It occurs at irregular intervals, and the last reversal was about 780,000 years ago.

Q7. Why does the Earth spin about its geographical axis instead of its magnetic axis?

Ans:

- The Earth spins about its geographical axis, which is the line joining its geographical poles.
- This rotation comes from the angular momentum the Earth got when it was formed.
- The magnetic axis is only due to the field made by molten iron in the core, and it has no role in the Earth's spin.

Q8. Why are the Earth's geographical and magnetic axes not coincident?

Ans:

- The geographical axis is decided by the Earth's rotation, while the magnetic axis is decided by the magnetic field made by the motion of molten iron in the core (dynamo effect).
- Since the two have different causes, they do not match.
- The magnetic poles are tilted by about 11.3° from the geographical poles, so a compass points 11.3° away from true north.

Q9. What is the difference between paramagnetic and ferromagnetic materials?

Ans:

- Paramagnetic materials: the spin and orbital motions of electrons do not fully cancel. In an external field they align with it and are weakly magnetized and weakly attracted, and they lose this magnetism when the field is removed. Examples: aluminium, sodium.
- Ferromagnetic materials: they have a net magnetism at the atomic level even without an external field. They are strongly magnetized and strongly attracted, and they keep their magnetism after the field is removed. Examples: iron, cobalt, nickel.

Q10. On what factors does the strength of an electromagnet (solenoid) depend?

Ans:

- The magnetic field of a solenoid is $B = \mu_0 nI$, where $n = N/L$.
- So the strength increases by increasing the number of turns per unit length (n) and by increasing the current (I).
- It also becomes much stronger by placing a soft iron core inside the coil, because soft iron magnetizes easily.

Q11. Draw magnetic field lines of two solenoids placed near each other: (i) facing same poles, (ii) facing opposite poles.

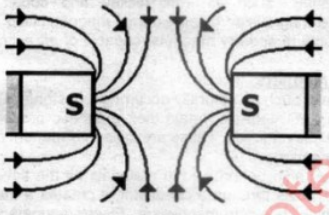
Ans:

- (i) Same poles: the field lines push away from each other, so a neutral (field free) zone forms between them.
- (ii) Opposite poles: the field lines go from the N-pole to the S-pole, so a strong field is formed between them.

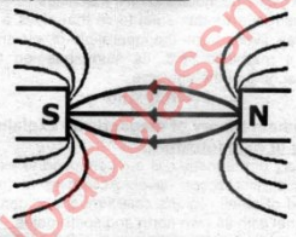
Q11. Draw magnetic field lines of two solenoids placed near each other:
(i) facing same poles to each other
(ii) facing opposite poles to each other

Ans:

(i) facing same poles to each other



(ii) facing opposite poles to each other



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